

Edelstein Effect - Relevant Scientific Sources

1. Edelstein Effect in Isotropic and Anisotropic Rashba Models

- **Authors:** Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, Roberta Citro
- **ArXiv ID:** 2503.20712
- **URL:** <https://arxiv.org/abs/2503.20712>
- **PDF:** <https://arxiv.org/pdf/2503.20712v1>
- **Found:** arXiv
- **Summary:** Investigates spin-to-charge conversion via the Edelstein effect in a 2D Rashba electron gas using the semiclassical Boltzmann approach. Analyzes magnetization arising from the direct Edelstein effect, considering anisotropic Rashba models. Studies dependence on effective mass, Fermi velocity, and other relevant parameters. **Most relevant for building a quantitative model.**

2. Boltzmann theory of the inverse Edelstein effect in a two-dimensional Rashba gas

- **Authors:** Irene Gaiardoni, Mattia Trama, Alfonso Maiellaro, Claudio Guarcello, Francesco Romeo, Roberta Citro
- **ArXiv ID:** 2601.02473
- **URL:** <https://arxiv.org/abs/2601.02473>
- **PDF:** <https://arxiv.org/pdf/2601.02473v1>
- **Found:** arXiv
- **Summary:** Derives analytical expressions for charge and spin currents within a semiclassical Boltzmann framework for Rashba 2DEG. Analyzes dependence on Rashba coupling strength, Fermi energy, and electric field. **Directly applicable for calculating magnetization magnitude and direction.**

3. Theory of the nonlinear Rashba-Edelstein effect

- **Authors:** Giovanni Vignale, I. V. Tokatly
- **ArXiv ID:** 1506.08330
- **URL:** <https://arxiv.org/abs/1506.08330>
- **PDF:** <https://arxiv.org/pdf/1506.08330v1>
- **Found:** arXiv
- **Summary:** Theoretical framework for current-driven spin polarization in 2D electron gas with Rashba spin-orbit coupling. Extends beyond linear response regime. Discusses drift velocity dependence and theoretical foundations. **Essential for understanding parameter dependencies (chirality, Fermi velocity).**

4. Spin accumulation at nonmagnetic interface induced by direct Rashba Edelstein effect

- **Authors:** Florent Auvray, Jorge Puebla, Mingran Xu, Bivas Rana, Daisuke Hashizume, Yoshichika Otani
- **ArXiv ID:** 1805.05523
- **URL:** <https://arxiv.org/abs/1805.05523>
- **PDF:** <https://arxiv.org/pdf/1805.05523v1>
- **Found:** arXiv
- **Summary:** Explains how Rashba effect generates spin polarization from charge current (Edelstein effect). Discusses momentum-dependent magnetic field coupling to electron angular momentum. **Provides fundamental understanding of direct Edelstein effect mechanism.**

5. Spin and orbital Edelstein effect in a bilayer system with Rashba interaction

- **Authors:** Sergio Leiva M., Jürgen Henk, Ingrid Mertig, Annika Johansson
- **ArXiv ID:** 2307.02872
- **URL:** <https://arxiv.org/abs/2307.02872>
- **PDF:** <https://arxiv.org/pdf/2307.02872v2>
- **Found:** arXiv
- **Summary:** Investigates spin and orbital Edelstein effects in systems with Rashba interaction. Discusses current-induced orbital magnetization and spin polarization. **Useful for understanding orbital contributions to magnetization.**

6. Spin Hall and Edelstein effects in a ballistic quantum dot with Rashba spin-orbit coupling

- **Authors:** Alfonso Maiellaro, Francesco Romeo, Mattia Trama, Jacopo Settino, Claudio Guarcello, Carmine Antonio Perroni, Pawel Wójcik, Bartłomiej Szafran, Daniela Stornaiuolo, Marco Salluzzo, Thomas Sand Jespersen, Nicolas Bergeal, Manuel Bibes, Roberta Citro
- **ArXiv ID:** 2602.02036
- **URL:** <https://arxiv.org/abs/2602.02036>
- **PDF:** <https://arxiv.org/pdf/2602.02036v1>
- **Found:** arXiv
- **Summary:** Studies charge-to-spin conversion in Rashba SOC systems. Identifies crossover from weak localization to weak antilocalization. **Relevant for understanding size-dependent effects in Rashba systems.**

7. Spin and orbital-to-charge conversion in noncentrosymmetric materials: Hall versus Rashba-Edelstein effects

- **Authors:** Diego Garcia Ovalle, Aurelien Manchon
- **ArXiv ID:** 2511.09511
- **URL:** <https://arxiv.org/abs/2511.09511>
- **PDF:** <https://arxiv.org/pdf/2511.09511v3>
- **Found:** arXiv
- **Summary:** Develops general formalism for spin- and orbital-to-charge conversion treating Hall effect and Rashba-Edelstein effect equally. Based on macroscopic theory. **Provides comprehensive theoretical framework for modeling.**

8. Slow-phonon control of spin Edelstein effect in Rashba d-wave altermagnets

- **Authors:** Mohsen Yarmohammadi, Jacob Linder, James K. Freericks
- **ArXiv ID:** 2510.02451
- **URL:** <https://arxiv.org/abs/2510.02451>
- **PDF:** <https://arxiv.org/pdf/2510.02451v2>
- **Found:** arXiv

- **Summary:** Investigates induced spin polarization in 2D d-wave altermagnets using Rashba coupling. Discusses intrinsic and externally induced spin polarizations. **Relevant for understanding parameter dependencies in Rashba models.**

9. Theory of spin pumping and inverse Rashba-Edelstein effect in a two-dimensional electron gas

- **Authors:** Masaki Yama, Mamoru Matsuo, Takeo Kato
- **ArXiv ID:** 2409.14267
- **URL:** <https://arxiv.org/abs/2409.14267>
- **PDF:** <https://arxiv.org/pdf/2409.14267v2>
- **Found:** arXiv
- **Summary:** Investigates spin transport in junction systems with Rashba and Dresselhaus SOC. Presents findings on spin pumping and inverse Edelstein effect. **Useful for understanding Rashba and Dresselhaus interplay.**

10. Enhanced Edelstein effect and interdimensional effects in an electron gas with Rashba spin-orbit coupling interface

- **Authors:** A. C. Zulkoskey, R. Dick, K. Tanaka
- **ArXiv ID:** 1912.01804
- **URL:** <https://arxiv.org/abs/1912.01804>
- **PDF:** <https://arxiv.org/pdf/1912.01804v1>
- **Found:** arXiv
- **Summary:** Examines bound-state and free-state contributions to density of states in 3D electron gas with 2D Rashba SOC interface. **Relevant for understanding interface effects on Edelstein effect.**

Key Parameters for Model Building:

Based on these sources, the following parameters should be considered when calculating the Edelstein effect for a Rashba fermion at the Gamma point:

- **Rashba coupling strength (α_R)**

- **Fermi velocity (v_F)**
- **Chirality of the Rashba bands**
- **Electric field magnitude and direction ($\vec{E}$)**
- **Fermi energy (E_F)**
- **Effective mass (m^*)**
- **Spin-orbit coupling strength**

The magnetization magnitude and direction depend on the applied electric field direction relative to the Rashba spin texture, with the spin polarization typically perpendicular to both the current direction and the spin-orbit field.